

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	22 June 2026
Name	Varsaa Vihasini S B
Project Name	Automated Network Request Management System
Maximum Marks	4 marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

1.1 The Core Operational Obstacle (Problem Statement)

The Velocity Bottleneck & Visibility Dark-Hole: Historically, corporate network provisioning at our organization relied on fragmented communication channels—adhoc emails, chat channels, and verbal confirmation loops. This manual pattern introduced a cascading set of operational problems: an unmonitored ticket fulfillment tail averaging 3.5 days, a high rate of configuration rework due to transcription errors when copying data between desks, and an absolute lack of progress visibility for end-users.

1.2 Strategic Disruptive Business Objective

To shift our service delivery model from a reactive ticket-logging pattern to a zero-touch, structured orchestration lifecycle natively powered by ServiceNow. The target state aims to establish a single source of intake that decreases total fulfillment latency by 80%, generates explicit compliance audit trails via automated data capturing, and establishes predictive field mechanics that maximize data integrity before an item is ever dispatched to engineering queues.

1.3 Matrix of Strategic Stakeholder Alignment

Stakeholder Persona	Enterprise Role	Core Expectations & Hidden Pain Points	Strategic Automation Impact
End Users (The Requesters)	Employees or functional teams requiring network service allocations through Service Portal exclusively.	Expectations: Simple, intuitive intake interfaces. Hidden Pain: Complete lack of horizontal tracking; unable to tell who is processing their request or where it sits.	Frictionless Intake: Instant form execution, clear placeholder guidance, and a transparent visual portal history track with real-time state alerts.

IT Admins (The System Architects)	Technical platform guardians managing instance health, core configurations, and workflow scaling.	Expectations: Standardized, upgrade-safe architectures. Hidden Pain: Over-customization and complex custom scripting bloat that breaks during native platform upgrade cycles.	Future-Proof Foundations: Implementation of native Variable Sets and out-of-the-box auto-population engine rules built for permanent PDI stability.
Network Fulfillment Team	Infrastructure engineering squads carrying out backend physical and logical network provisioning adjustments.	Expectations: Deterministic, structured, and complete input profiles. Hidden Pain: Wasting hours tracking down missing location attributes or device parameters across disparate email chains.	Deterministic Execution: Automated tasking layers (u_network_task) tracking directly against centralized custom data inventories (u_network_database).
Approvers (The Managers)	Line-of-business supervisors, departmental heads, and corporate infrastructure compliance gatekeepers.	Expectations: Fast, low-friction policy enforcement checkpoints. Hidden Pain: Hunting through unorganized email loops for approvals that fail to generate structured audit trails.	Systematic Accountability: Dynamic manager field dot-walking logic combined with absolute, immutable lifecycle transaction histories for auditing.

Step-2: Brainstorm, Idea Listing and Grouping

During our collaborative cross-functional design sprints, potential infrastructure solutions were grouped into three focus pillars: Intake Layer Innovation, Workflow Automation Intelligence, and Fulfillment Integrity.

Pillar A: Intake Layer Innovation (UI/UX Engineering)

- **Idea A1 [Standard Dynamic Forms]:** Utilize broad, free-form text elements for parameter collection to allow free-form user tracking. (Status: Rejected due to

high risk of user syntax errors, formatting anomalies, and un-executable configuration parameters).

- **Idea A2 [Contextual Identity Binding]:** Implement a runtime user context hook mapping the logged-in portal account directly to target corporate structures (sys_user table), instantly auto-filling name, corporate email, and corporate telephone numbers. (Status: Selected - Implemented via reference variables).
- **Idea A3 [Inline Descriptive Micro-Annotations]:** Embed clear gray instructional placeholder prompts directly within empty form blocks to enforce strict formatting boundaries without cluttering visual layout real estate. (Status: Selected - Implemented via variable annotations).

Pillar B: Workflow Automation Intelligence (Process Engine Architecture)

- **Idea B1 [Linear Hierarchical Approvals]:** Route tracking records sequentially step-by-step up the entire regional organizational structural tier. (Status: Rejected as it introduces extensive administrative gridlock and artificially balloons resolution metrics).
- **Idea B2 [Dynamic Field-Pill Dot-Walking]:** Utilize Flow Designer parameter models to dynamically extract the explicit corporate manager record of the user specified in the 'Opened on behalf of' field, isolating backend engine execution from hardcoded sys_ids. (Status: Selected - Implemented via Ask For Approval action paths).
- **Idea B3 [Branching Risk Exceptions]:** Establish execution splits checking for non-standard hardware definitions or high-sensitivity relocations to alert the specialized Information Security infrastructure team automatically. (Status: Selected - Implemented via internal dynamic If-Else logical checks).

Pillar C: Fulfillment Integrity (Backend Execution Architecture)

- **Idea C1 [Email-Driven Dispatching]:** Rely entirely on standard platform email alerts notifying infrastructure engineers to log in and find open items manually. (Status: Rejected as it fails to provide formal tracking status and leads to lost tickets).
- **Idea C2 [Transactional Record Mirroring]:** Programmatically parse verified portal variables and construct mirrored transaction tracking records inside an isolated custom tracking ledger (u_network_database) while concurrently managing child tasks (u_network_task). (Status: Selected - Implemented via ServiceNow Core Create Record actions).
- **Idea C3 [API-Driven Direct Orchestration Hubs]:** Leverage ServiceNow Integration Hub spokes to broadcast configuration payloads directly into external infrastructure controller environments (e.g., Ansible Playbooks / Cisco DNA Center). (Status: Deferred to Future Scale Roadmap due to scope limitations inside baseline PDI environments).

Step-3: Idea Prioritization

To optimize execution, isolate high-value deliverables, and prevent resource bleed during development, all brainstormed concepts were measured using an Impact vs. Implementation Effort Matrix framework.

☐ QUICK WINS (High Impact / Low Effort) • Contextual Identity Binding • Variable Set Modularization • Dynamic Manager Dot-Walking • Inline Placeholder Guidance	☐ STRATEGIC CORNERSTONES (High Impact / High Effort) • Custom Ledger Schema Creation • Multi-Tier Record Mirroring • Branching Tasking Topologies
☐ DEFERRED VALUE (Low Impact / Low Effort) • Free-Form Text Entry Ingestion • Linear Hierarchy Approval Paths	☐ FUTURE ROADMAP (High Impact / Very High Effort) • Zero-Touch Spoke Orchestration with External Hardware Automation APIs

3.1 Strategic Selection & Justification Analysis

1. The Quick Wins (High Impact / Low Effort — Implemented Immediately):

Justification: Creating the standardized 'Requester information' Variable Set combined with dependent Auto-populate actions provides massive platform value with low design overhead. It permanently eliminates standard data-entry typos by automatically pulling validated production data records directly from primary system identity tables, radically improving user satisfaction from the first layout interaction.

2. The Strategic Cornerstones (High Impact / High Effort — Core Development Track):

Justification: Separating the portal's volatile catalog item runtime data from long-term active engineering tracking fields requires a distinct customized database architecture backend (u_network_database and u_network_task schemas). While this layer introduces database structural design complexity and requires explicit field mapping loops within Flow Designer, it guarantees absolute database records isolation and protects transactional audit integrity.

3. The Future Expansion Roadmap (High Impact / Very High Effort — Scheduled for Next Release):

Justification: Transitioning our automated task closures into actual direct infrastructure hardware configuration deployments via API layer connections represents our ultimate operational targets. By constructing modular engineering child execution tasks inside our core workflow right now, we have intentionally built the structural data endpoints needed to seamlessly support integration hub scripts during our next sprint cycle without re-architecting the system.